



## First Semester 2026-27

### Course Handout Part II

Date: 3<sup>rd</sup> August 2026

In addition to part I (General Handout for all courses appended to the Timetable), this portion gives further specific details regarding the course.

**Course Number** : CHEM F213

**Course Title** : Physical Chemistry-II

**Instructor-in-charge** : Prof. Avik Kumar Pati

- Course description:** Origin of quantum theory-black body radiation, line spectra, photoelectric effect; wave particle duality; wave equation: normal modes, superposition; postulates of quantum mechanics, time dependence, Hermitian operators, commutator; Schrödinger equation, operators, observables, solution for particle in a box, normalization, variance, momentum; harmonic oscillator, vibrational spectroscopy; rigid rotor, angular momentum, rotational spectroscopy; Hydrogen atom-orbitals, effect of magnetic field; Variation method, variation theorem, secular determinants; Many electron atoms and molecules; Born Oppenheimer approximation, VB Theory,  $H_2$  in VB, Coulomb, ex-change, overlap integrals states of  $H_2$ ; antisymmetric wavefunctions two electron systems, Slater determinants, HF method; SCF method; term symbols and spectra, configuration, state, Hund's rules, atomic spectra, spin orbit interaction; basic MO theory, homonuclear diatomics- $N_2$ ,  $O_2$ , SCF-LCAO-MO, molecular term symbols; HMO theory- $\pi$  electron approximation, conjugated, cyclic systems.
- Scope and Objective:** This is the second of the four Physical Chemistry courses designed for the integrated M.Sc. Chemistry Programme. The course introduces quantum mechanics and application of quantum mechanics to electronic structure of atoms/molecules, chemical bonding, and spectroscopy. At the end of the course the student will be able to understand fundamentals of quantum chemistry and can apply quantum mechanics to electronic structure calculation of many electrons atoms and into molecular spectroscopy.
- Course learning outcomes:**
  - Demonstrate the need of quantum mechanics to explain atomic and molecular phenomenon.
  - Apply concepts of quantum mechanics to simple and exactly solvable physical systems.
  - Explain bonding theories and their application to simple molecular systems.
  - Illustrate various approximation methods using quantum mechanics.
  - Formulate SCF methods for many electrons system which is the basic of any electronic structure calculation for geometry optimization and energy calculation.
  - Application of quantum mechanics into molecular spectroscopy.

#### 4. Text books:

Quantum Chemistry, Donald A McQuarrie, University Science Books (First Indian Edition 2003, Viva Books Private Limited)

#### 5. Reference Books (RB):

**RB-1:** *Quantum Chemistry*, Ira N. Levine, 7<sup>th</sup> Edition, PHI, 2014,

**RB-2:** *Quantum Mechanics*, Bransden and Joachain, 2<sup>nd</sup> Edition, Pearson.

**RB-3:** *Atkins' Physical Chemistry*, 9<sup>th</sup> Edition, Peter Atkins and Julio De Paula, Oxford University Press.

**RB-4:** *Introduction to Quantum Mechanics with Applications to Chemistry*, Linus Pauling and E. Bright Wilson, Jr., Dover Publications, Inc, New York.

**RB-5:** *Elements of Quantum Mechanics*, Michael D. Fayer, Oxford University Press, New York, 2001.

#### 6. Course Plan:

| Week no. | Module                            | Lect. no. | Learning Objective/Outcome                                                                                                                                                                                                                                                                                                                                                                    |
|----------|-----------------------------------|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1-3      | 1. Development of Quantum Theory  | [1-2]     | <b>Failure of classical physics and origin of quantum theory.</b><br><b>Objective:</b> Blackbody radiation, Photoelectric Effect, Atomic Vibrations in Crystals, Line Spectra and Bohr Model of H Atom, de Broglie's postulate, Heisenberg uncertainty principle.<br><b>Outcome:</b> Recognize the necessity of quantum theory.<br>[TB1 §1.1-1.4]                                             |
|          |                                   | [3]       | <b>The wave equation</b><br><b>Objective:</b> Motion of vibrating string, separation of variables, normal modes, superposition, Fourier series.<br><b>Outcome:</b> Recognize mathematical background for quantum theory.<br>[TB1 §2.1-2.5]                                                                                                                                                    |
|          |                                   | [4-8]     | <b>Postulates of quantum mechanics</b><br><b>Objective:</b> Wave function, Schrödinger equation, operators and observables, eigenvalue problem, time evaluation and stationary states, uncertainty, measurement, and superposition of states.<br><b>Outcome:</b> Consolidate new concepts to be used in quantum mechanics<br>[TB1 § 3.1 – 3.4, 3.7, 3.8, 3.11, 4.1 – 4.9]<br>[RB2 § 7.1-7.10] |
| 4-8      | 2. Some exactly solvable problems | [9-10]    | <b>Particle in a Box (PIB)</b><br><b>Objective:</b> Bound states, Zero-point energy, symmetry, superposition states, and degeneracy in two- and three-dimensions.<br><b>Outcome:</b> Theorize quantization of states and zero point energy in very simple systems, like, PIB.<br>[TB § 3.4 – 3.11, 6.1 – 6.2]                                                                                 |



|       |                          |         |                                                                                                                                                                                                                                                                                                            |
|-------|--------------------------|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|       |                          | [11-12] | <b>Finite potential wells and barriers</b><br><b>Objective:</b> Bound states in wells, probability current, reflections and tunneling.<br><b>Outcome:</b> Recognize working principle of scanning tunneling microscopy.<br>[RB-2 § 2.5; RB-3 § 8.3]                                                        |
|       |                          | [13-15] | <b>Harmonic Oscillator</b><br><b>Objective:</b> Eigen states for simple harmonic oscillator, zero-point energy, molecular vibration, dissociation energy, and anharmonicity.<br><b>Outcome:</b> Define vibrational spectroscopy of molecules.<br>[TB § 5.1 – 5.13, RB-4 § 11a-11c]                         |
|       |                          | [16-18] | <b>Angular momentum and rigid rotor</b><br><b>Objectives:</b> Energy levels, operator for angular momentum, commutation relation, wavefunction, and molecular rotation.<br><b>Outcome:</b> Define rigid rotator as model for rotating diatomic molecules.<br>[TB § 6.3 – 6.7, 6.10, RB-4 § 18-21]          |
|       |                          | [19-20] | <b>The hydrogen atom</b><br><b>Objectives:</b> Energy levels, wavefunction – angular and radial parts, orbitals, and effect of magnetic field on hydrogen atom.<br><b>Outcome:</b> Review atomic orbital picture of H-atom from quantum mechanics.<br>[TB § 6.8 – 6.11]                                    |
| 8-9   | 3. Approximation methods | [21-23] | <b>Variation method</b><br><b>Objectives:</b> Schrödinger equation for He-atom cannot be solved exactly, variation theorem, linear variation method.<br><b>Outcome:</b> Evaluate the upper bound to the ground state energy of a system.<br>[TB § 6.12, 7.3 – 7.7, 8.1, 8.2]                               |
|       |                          | [24-25] | <b>Perturbation theory for stationary state</b><br><b>Objectives:</b> Systematic correction of wavefunction and energies of non-degenerate states.<br><b>Outcome:</b> Calculate ground state energy of various systems from the unperturbed state of the system.<br>[TB § 7.1, 7.2, 8.2; RB-2 § 9.1 – 9.7] |
| 10-11 | 4. Many electron         | [26]    | <b>Many electron wavefunction</b>                                                                                                                                                                                                                                                                          |

|       |              |         |                                                                                                                                                                                                                                                                                                                             |
|-------|--------------|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|       | atoms        |         | <p><b>Objectives:</b> Systems of identical particles, spin and permutation symmetry, Pauli principle, and Slater determinants.</p> <p><b>Outcome:</b> Define spin as another coordinate.</p> <p>[TB § 8.4 – 8.6]</p>                                                                                                        |
|       |              | [27-29] | <p><b>SCF method</b></p> <p><b>Objectives:</b> Hartree and Hartree-Fock methods, electronic structure calculations, comparison with experimental data, periodicity.</p> <p><b>Outcome:</b> Recognize theoretical concepts behind electronic structure calculations of polyatomic molecules.</p> <p>[TB § 8.3, 8.7, 8.8]</p> |
|       |              | [30]    | <p><b>Atomic terms and spectra</b></p> <p><b>Objectives:</b> Addition of angular momenta, spin-orbit interaction, absorption and emission spectra of atoms, selection rules.</p> <p><b>Outcome:</b> Examine allowed and forbidden transition in atoms.</p> <p>[TB § 8.9-8.12]</p>                                           |
| 11-13 | 5. Molecules | [31]    | <p><b>Born-Oppenheimer approximation</b></p> <p><b>Objective:</b> Separation of nuclear and electronic motion.</p> <p><b>Outcome:</b> Express molecular wavefunction as product of nuclear and electronic wavefunctions.</p> <p>[TB § 9.1]</p>                                                                              |
|       |              | [32-33] | <p><b>Valence Bond Theory – H<sub>2</sub></b></p> <p><b>Objective:</b> Localized electron pair bonds.</p> <p><b>Outcome:</b> Demonstrate successful description of chemical bond.</p> <p>[TB § 9.2 – 9.5]</p>                                                                                                               |
|       |              | [34-35] | <p><b>Molecular Orbital Theory – H<sub>2</sub><sup>+</sup>, H<sub>2</sub></b></p> <p><b>Objective:</b> Linear combination of atomic orbitals, comparison to valence bond picture.</p> <p><b>Outcome:</b> Illustrate application of molecular orbital theory to diatomic molecules.</p> <p>[TB § 9.6 – 9.8]</p>              |
|       |              | [36-37] | <p><b>Homonuclear diatomic molecules</b></p> <p><b>Objective:</b> Molecular electronic configuration, Pauli principle, SCF-LCAO-MO wavefunctions, and molecular terms.</p> <p><b>Outcome:</b> Compare experimental observations along with theoretical prediction for diatomic molecules.</p> <p>[TB § 9.9 – 10.5]</p>      |



|       |                           |      |                                                                                                                                                                                                                                                                                                                                                                                                            |
|-------|---------------------------|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|       |                           | [38] | <b>Hückel MO theory</b><br><i><b>Objective:</b></i> $\pi$ -electron approximation for conjugated systems, energies and delocalization, atomic charge distribution and bond order.<br><i><b>Outcome:</b></i> Explore quantum chemical approximation of aromatic systems.<br>[TB § 9.21 – 9.24]                                                                                                              |
| 14-15 | 6. Molecular spectroscopy | [39] | <b>Electromagnetic radiation and its interaction with molecules</b><br><i><b>Objective:</b></i> Perturbation theory of degenerate energy levels, time dependent perturbation theory, interaction of radiation and matter, absorption, emission.<br><i><b>Outcome:</b></i> Recognize quantum-mechanical approach for spectroscopy.<br>[RB-2 § 9.5, 9.9, 9.10]                                               |
|       |                           | [40] | <b>Rotational and vibrational spectra</b><br><i><b>Objective:</b></i> Different regions of electromagnetic spectrum, molecular rotation, molecular vibration, normal modes, rotational transitions accompany vibrational transitions, Boltzmann population distribution.<br><i><b>Outcome:</b></i> Explain rotational and vibrational spectroscopy.<br>[TB § 10.1 – 10.10]                                 |
|       |                           | [41] | <b>Electronic spectra</b><br><i><b>Objective:</b></i> Electronic states, transitions, vibrational and rotational information, intensity pattern.<br><i><b>Outcome:</b></i> Recognize fundamentals of electronic spectroscopy.<br>[TB § 10.11 – 10.13]                                                                                                                                                      |
|       |                           | [42] | <b>Selection rules and Franck-Condon principle</b><br><i><b>Objective:</b></i> Overlap integrals, nuclear motion can be factored approximately into a rotational part and vibrational part, selection rule in rigid rotor, selection rule in harmonic oscillator, selection rule in electronic spectroscopy.<br><i><b>Outcome:</b></i> Formulate allowed and forbidden transition.<br>[TB § 10.14 – 10.18] |

## 7. Evaluation Scheme:

| Component <sup>β</sup>                                   | Duration           | Weightage (%)<br>(Total Marks 200) | Date & Time                      | Remarks                                                  |
|----------------------------------------------------------|--------------------|------------------------------------|----------------------------------|----------------------------------------------------------|
| Continuous Evaluation-Type I<br>(Quiz) <sup>§</sup>      | To Be<br>Announced | 20% (40 marks)                     | To Be<br>Announced               | Closed Book                                              |
| Continuous Evaluation –Type<br>II<br>(Quiz) <sup>*</sup> | To Be<br>Announced | 10% (20 marks)                     | To Be<br>Announced               | Closed Book                                              |
| Mid-Semester Test                                        | 90 Minutes         | 30% (60 marks)                     | As per the<br>AUGSD<br>timetable | Closed Book                                              |
| Comprehensive Examination                                | 3 hours            | 40% (80 marks)                     | As per the<br>AUGSD<br>timetable | Partially Closed and<br>Partially Open Book <sup>α</sup> |

<sup>§</sup> Continuous evaluation- Type I: A total of THREE quizzes (each of 20 marks) will be conducted under continuous evaluation. **Best TWO** will be considered for final evaluation (*i.e.*, ONE buffer).

<sup>\*</sup> Continuous evaluation- Type II: This evaluation is linked to ATTENDANCE. A total of THREE quizzes (each of 10 marks) will be conducted under continuous evaluation. **Best TWO** will be considered for final evaluation (*i.e.*, ONE buffer). **This Type-II quiz marks (10%) will be considered for final grading only if the student maintains a minimum of 50% attendance.** This requirement applies to students with ID numbers 2024B2XXXX and 2025B2XXXX.

<sup>α</sup> Only textbooks, reference books, class/tutorial hand-written notes, and course material (if any provided) will be allowed in the open book examination.

<sup>β</sup> Only scientific non-programmable calculators are allowed during all the examinations, and you must bring your own calculator.

## 8. Minimum criterion for awarding valid grades:

Where a student remains registered in a course but provides the instructor with inadequate opportunity for evaluation, such as by being absent from continuous assessments (Type-I), mid-semester examinations, or comprehensive examinations, or by appearing for these evaluations merely for formality without meaningfully attempting the tasks, including submitting a blank answer script; the student will be reported as **NC (Not Cleared)**.

Furthermore, to clear the course, a student must obtain **at least 30% of the average marks of the top 10 performers in the class, or 40% of the class median marks, whichever is lower.** Any student whose performance falls below this prescribed standard may be assigned to **NC**. This will be applicable to students of ID No. 2024B2XXXX and 2025B2XXXX.



# Birla Institute of Technology & Science, Pilani

Pilani Campus

**ACADEMIC— UNDER GRADUATE STUDIES DIVISION**

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9. **Chamber Consultation Hour:** Wednesday 5 PM – 05:50 PM
10. **Notices:** Notices and course related announcements will be communicated to the students through Nalanda and in the classes as well.
11. **Make-up Policy:** Make-up will be considered for genuine cases as per the Institute rules against an application by the student with proper documents. See Part I for details.

**Avik Kumar Pati**

**Instructor-in-charge**

**Course No. CHEM F213**